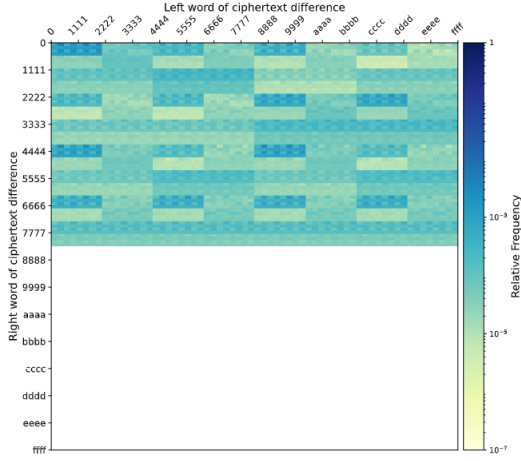
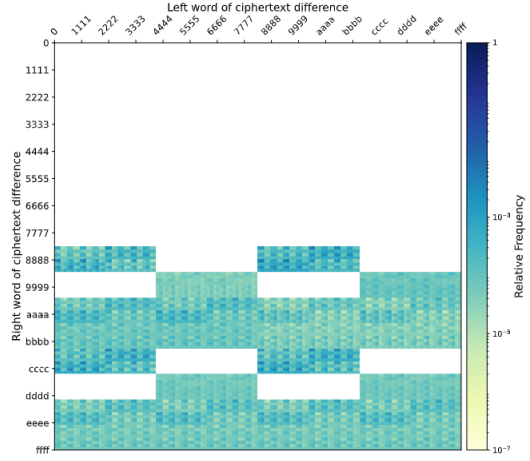


We believe that using two distinct input differences may increase the separation of differential distributions, particularly when these distributions cluster in different regions. This could help the neural network better distinguish ciphertext pairs and potentially increase the number of distinguishable rounds. To provide a preliminary validation, we select five pairs of input differences (δ_1, δ_2) from Figure 8 of our paper and evaluate them using a ResNet-based model. The results show a negative correlation between training accuracy and the degree of similarity between differential distributions, providing initial support for our hypothesis.

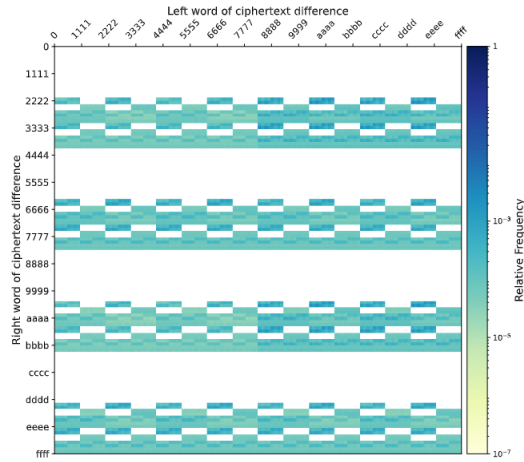


(a) $0x00000001$, 0.748

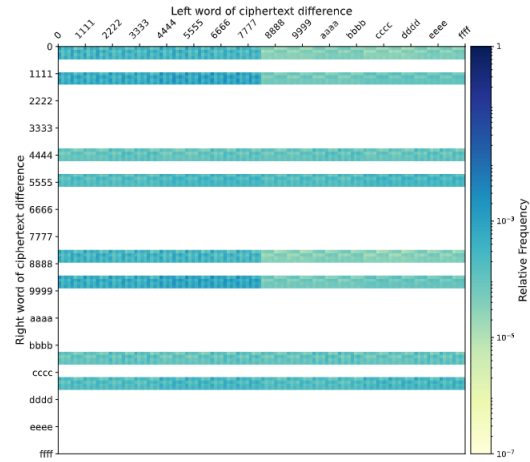


(b) $0x00000080$, 0.749

$(\delta_1 = 0x00000001, \delta_2 = 0x00000080)$, Accuracy=1.000

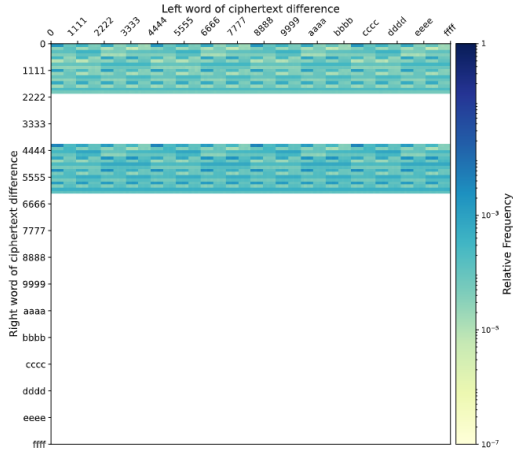


(c) $0x00000020$, 0.750

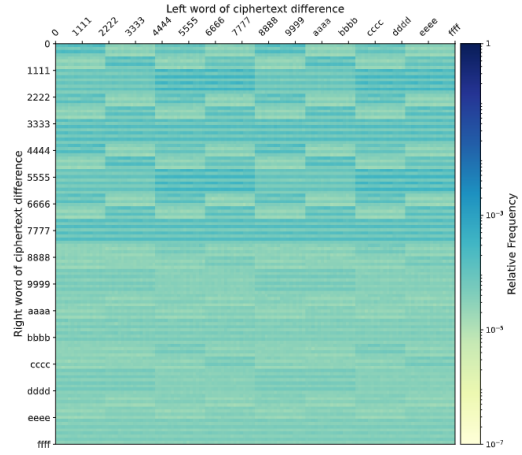


(d) $0x00001000$, 0.743

$(\delta_1 = 0x00000020, \delta_2 = 0x00001000)$, Accuracy=1.000

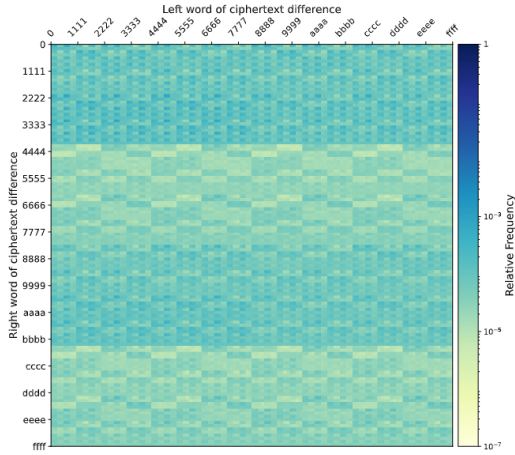


(c) 0x00004000, 0.749

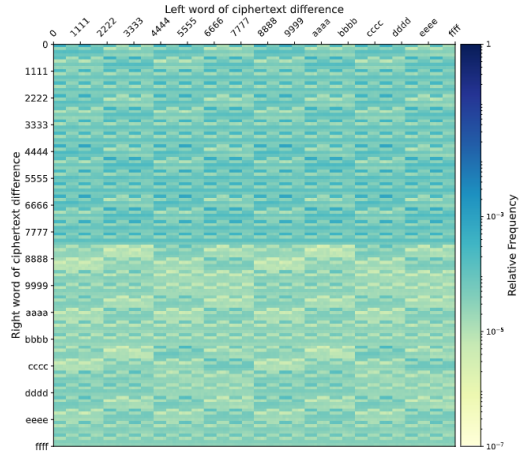


(d) 0x00040000, 0.610

$(\delta_1 = 0x00004000, \delta_2 = 0x00040000), \text{Accuracy}=0.982$

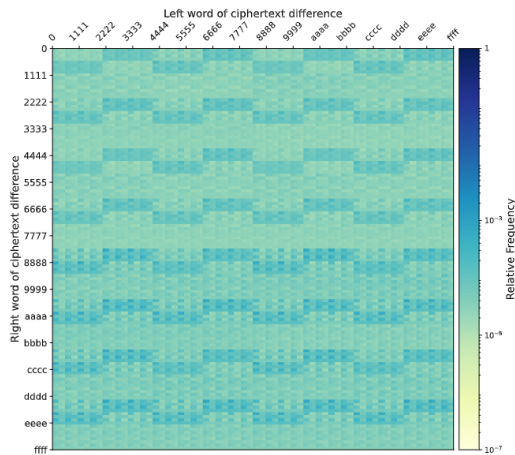


(e) 0x08000000, 0.611

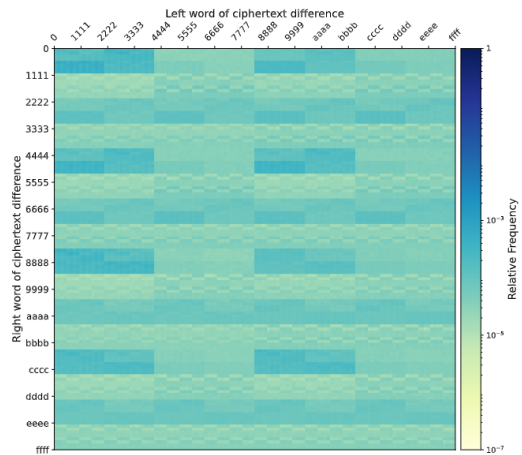


(f) 0x10000000, 0.609

$(\delta_1 = 0x08000000, \delta_2 = 0x10000000), \text{Accuracy}=0.977$



(a) 0x00200000, 0.629



(b) 0x02000000, 0.611

$(\delta_1 = 0x00200000, \delta_2 = 0x02000000), \text{Accuracy}=0.949$